

PRODUCT REQUIREMENTS DOCUMENT

WordGraph

Visual Vocabulary Explorer

"Obsidian meets a thesaurus. No gamification. No streaks. Just words."

VERSION

v2.0

STATUS

Living document

AUDIENCE

Engineering · Design ·
Product

1 Product Vision

WordGraph is a calm, local-first vocabulary tool for people who care about words. It lets you search any English word, explore how it connects to synonyms, antonyms, and related words in an interactive graph, and build a personal library of words worth keeping — all without an account, a server, or the internet after first load.

2 Target Users

Persona	Description	Core need
The Writer	Novelist, copywriter, journalist	Find the exact word; avoid repetition
The Learner	ESL student, self-taught reader	Understand a word deeply, not just its definition
The Thinker	Researcher, academic, knowledge worker	Build a personal vocabulary corpus; export for reference
The Browser	Curious reader encountering an unfamiliar word	Quick lookup that doesn't interrupt flow

3 Design Principles

1. **Typography first** — the word is the UI. No coloured bubbles, no icons fighting for attention.
2. **Local-first** — your data lives on your device. No account, no cloud sync, no data collection.
3. **Calm** — no streaks, no badges, no notifications. The app recedes; the words stand forward.
4. **Fast by default** — instant for cached/static words; graceful loading for fresh lookups.
5. **Accessible** — WCAG 2.1 AA; full keyboard navigation; reduced-motion support.

4 Tech Stack

FRONTEND

Layer	Choice	Reason
Framework	React 18 + TypeScript	Component model, ecosystem, type safety
Build tool	Vite 7	Fast HMR, ESM-first, easy to extend for extension build
Routing	Wouter	Tiny (2 KB), hook-based, no context provider boilerplate
Animation	Framer Motion	<code>useReducedMotion</code> -aware, declarative graph transitions
Styling	CSS custom properties + utility classes	Design system token-driven, no runtime CSS-in-JS
Icons	Lucide React	Consistent, tree-shaken

DESIGN SYSTEM — @WORKSPACE/WORDGRAPH-DESIGN-SYSTEM

Area	Approach
Tokens	<code>tokens.json</code> → generated <code>styles.css</code> (CSS custom properties)
Colour	Two palettes: <code>light</code> / <code>dark</code> . Graph semantic: synonym (muted green), antonym (muted red), related (muted blue)
Typography	Inter, type scale via <code>--font-*</code> variables
Components	Shared React components consumed by the web app

PERSISTENCE

Layer	Technology	Purpose
Word graph	Static <code>WORD_DB</code> (TypeScript)	12 seed words, always authoritative
Dictionary cache	Dexie.js / IndexedDB	Cache API responses for 7 days; works offline after first load
User library	Dexie.js / IndexedDB	Saved words, collections, tags, notes — never leaves the device

DICTIONARY

Attribute	Detail
Primary source	Free Dictionary API — open, no key, ~100k common English words
Endpoint	<code>GET https://api.dictionaryapi.dev/api/v2/entries/en/{word}</code>
Fallback source	Datamuse API — fills gaps for abstract/rare words (synonyms, antonyms, related)
Offline behaviour	IndexedDB cache → not-found state (no crash)
Coverage gap	Abstract/philosophical words, neologisms, proper nouns

TESTING & MONOREPO

Tool	Role
Playwright	E2E: save, tag, notes, collection, export fidelity, import idempotence — 6 tests, all passing
pnpm workspaces	Monorepo package management, cross-package deps
TypeScript project refs	Incremental builds

5 Information Architecture

/	→ Home (search bar, recent words)
/explore/:word	→ Explore (graph + word info panel)
/library	→ Library (all saved words)
/library/collections/:id	→ Collection view
/library/tags/:tag	→ Tag view
/settings	→ Export / import

6 Feature Inventory

6.1 SEARCH & DISCOVERY

Feature	Status	Notes
Search any English word	☒ Shipped	Free Dictionary API + IndexedDB cache
Recent words on home screen	☒ Shipped	Stored in sessionStorage
Autocomplete while typing	☐ Proposed	Match cached + static words; Datamuse suggestions
Global search from every page	☐ Proposed	Not just home screen

6.2 GRAPH

Feature	Status	Notes
Interactive SVG graph	📦 Shipped	Typographic nodes, pan support
Synonym / antonym / related lanes	📦 Shipped	Colour-coded edges
Click node → explore that word	📦 Shipped	Pushes new URL
Legend overlay	📦 Shipped	Top-left of graph SVG, always visible
Exploration breadcrumb	📦 Shipped	Session path: Pragmatic → Realistic → ...
Multi-word phrase filtering	📦 Proposed	Filter API nodes containing spaces

6.3 WORD INFORMATION PANEL

Feature	Status	Notes
Definition, part of speech, IPA	📦 Shipped	
Commonness badge	📦 Shipped	Common / uncommon heuristic
Examples	📦 Shipped	Up to 2 from API
Memory trick & usage notes	📦 Shipped	Static DB only
Audio pronunciation	📦 Proposed	Free Dictionary API includes audio URLs

6.4 LIBRARY & COLLECTIONS

Feature	Status	Notes
Save / unsave word	📦 Shipped	
Personal notes	📦 Shipped	Debounced auto-save
Tags	📦 Shipped	Multi-tag, inline add / remove
Collections	📦 Shipped	CRUD, word membership
Library graph	📦 Shipped	Graph of all saved words
Filter by collection / tag	📦 Shipped	Sidebar navigation

6.5 DATA OWNERSHIP

Feature	Status	Notes
Export JSON	☒ Shipped	Full snapshot: words, tags, notes, collections, memberships
Export Markdown	☒ Shipped	Human-readable
Export CSV	☒ Shipped	Spreadsheet-compatible
Import JSON	☒ Shipped	Merge semantics: no data loss, deduplicated
E2E data-fidelity tests	☒ Shipped	6 Playwright tests — all passing

6.6 UX

Feature	Status	Notes
Light / dark mode	☒ Shipped	System preference + manual toggle
Reduced motion	☒ Shipped	Respects <code>prefers-reduced-motion</code>
Responsive layout	☒ Shipped	Desktop: side-by-side; mobile: stacked
Keyboard navigation	☒ Shipped	Focus rings on all interactive elements
ARIA labels	☒ Shipped	Graph nodes, edges, legend, all buttons

7 Data Model

INDEXEDDB SCHEMA (DEXIE)

```
// Database: "wordgraph"

savedWords          // PK: word (string)
  word              string
  savedAt           number   (Unix ms)
  notes             string
  tags              string[]

collections          // PK: ++id (auto-increment)
  id                number
  name              string
  createdAt          number

wordCollections      // Compound index: [word+collectionId]
  word              string
  collectionId       number

wordApiCache         // PK: word (string)
  word              string
  data              string   // JSON | '__NOT_FOUND__'
  cachedAt          number   // TTL: 7 days

recentWords          // sessionStorage (not Dexie)
```

EXPORT SNAPSHOT SCHEMA (V1)

```
{
  version:          1,
  exportedAt:        string,          // ISO-8601
  savedWords:        SavedWord[],
  collections:       Collection[],
  wordCollections:   WordCollection[]
}
```

8 API Dependency

Attribute	Free Dictionary API	Datamuse API (fallback)
URL	<code>api.dictionaryapi.dev/api/v2/entries/en/{word}</code>	<code>api.datamuse.com/words?rel_syn={word}</code> etc.
Auth	None	None
Rate limits	Undocumented; no SLA	Undocumented; generous for reasonable use
Coverage gaps	Abstract/philosophical words, neologisms, proper nouns	Stronger on rare words; weaker on definitions
Offline	Falls back to IndexedDB cache	Falls back to IndexedDB cache

9 Non-Functional Requirements

Requirement	Target
First contentful paint (cached word)	< 300 ms
First contentful paint (fresh API word)	< 1.5 s
Initial JS bundle (gzipped)	< 200 KB
Offline capability	Full — after first word load
Data privacy	Zero telemetry; no network calls except dictionary lookups
Browser support	Chrome 114+, Firefox 115+, Safari 17+, Edge 114+
Accessibility	WCAG 2.1 AA

10 Roadmap

V1 — SHIPPED

Search any word → explore graph → save to library → manage collections & tags → export / import.

V1.1 — NEXT

- Audio pronunciation playback (Free Dictionary API has audio URLs)
- Filter multi-word graph nodes from Datamuse / dictionary responses
- Global search from every page + autocomplete suggestions while typing

V1.2 — LATER

- Keyboard shortcut to save current word
- Shareable word links (word encoded in URL, no server needed)
- Richer related-word graph via Datamuse `rel_jja` / `rel_trg` endpoints

V2 — STRETCH

- Browser extension — right-click any word on any page (see *Browser Extension Feasibility* doc)
- PWA / install-to-home-screen
- Spaced-repetition review mode (opt-in, no gamification aesthetic)
- Obsidian vault export

11 Metrics

WordGraph is intentionally zero-telemetry. Success is measured qualitatively:

- User-reported library exports growing over time
- Extension install count once shipped
- Community contributions / GitHub stars

12 Out of Scope (v1)

- User accounts / cloud sync

- Social features (sharing, leaderboards)
- Mobile native app (evaluate after browser extension)
- AI-generated definitions
- Monetisation — free and open